Refocusing of Rotating Ships in Spaceborne SAR Imagery Based on NLCS Principle

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***Abstract*— Rotating ships usually suffer from complicated defocusing in high-resolution spaceborne synthetic aperture radar (SAR) imagery due to the nonuniform motions, which severely restricts the classification and interpretation of ships in SAR applications. In this article, a novel refocusing method is proposed to improve the imaging quality of rotating ships by postprocessing SAR imagery. We first attempt to utilize the nonlinear chirp scaling (NLCS) principle to correct space-variant phase errors induced by target rotations. Specifically, an azimuth inverse compression approach without zero-padding operation is designed to recover the azimuth frequency modulation (FM) of the target image data. Then, a modified NLCS operation is derived for space-variant phase error correction by introducing a high-order perturbation function. An objective function based on image quality is also constructed to estimate the optimal perturbation parameters. Finally, simulated data and real spaceborne SAR data are processed to demonstrate the good performance of the proposed method in comparison with other state-of-the-art techniques.**

***Index Terms*— Nonlinear chirp scaling (NLCS), refocusing, rotational motions, ship target, synthetic aperture radar (SAR).**

I. Introduction

SPACEBORNE synthetic aperture radar (SAR) has been an effective means of remote sensing observation with all-weather and all-time imaging capabilities [1]. One typical SAR application is marine surveillance, for which high-resolution SAR images with wide-area coverage of the sea are formed by SAR sensors from the satellite. However, moving ships, which are targets of interest in both military and civil fields, usually appear blurred in SAR imagery because of their motions during the observation time [2]. Unknown phase errors introduced by motions of the targets will cause a mismatch of the SAR image formation algorithm designed for stationary scenes, leading to degradation of the target's image quality, such as defocusing, distortion, or displacement. This severely restricts the interpretation of the target. Therefore, refocusing ship targets in SAR imagery via postprocessing has brought much attention.

Many autofocus algorithms have been proposed to estimate and correct phase errors caused by ship's translational motions. Scatterer-based methods, such as phase gradient autofocus (PGA) algorithms [3], [4], [5], [6], estimate phase errors via strong point scatterers. Metric-based methods realize motion compensation by optimizing SAR image quality, such as contrast, entropy, and sharpness [7], [8], [9], [10]. Generally, target's motion along the azimuth direction leads to a residual chirp and the motion along the range direction causes a position shift. Therefore, some researchers propose to estimate phase errors of moving targets by time–frequency analysis [2], [11], [12]. The above algorithms assume that the motions of all point scatterers belonging to a ship are identical. Therefore, phase error compensation is conducted in the azimuth frequency domain to make all point scatterers focused simultaneously. These approaches work well in the case that the observation time is much less than 1 s, in which the ship's sailing velocity is the main source of the motion. However, with the improvement of the SAR resolution, the length of synthetic aperture time increases, and ship rotations (i.e., roll, yaw, and pitch) induced by sea breezes and waves can no longer be ignored [13]. The effects of ship rotations on SAR imaging are analyzed in [14], [15], and [16]. The nonuniform motions will introduce space-variant phase errors to the target's signals, which are difficult to be corrected by traditional refocusing methods mentioned above.

In [17], [18], and [19], inverse SAR (ISAR) technique is first utilized to overcome this difficulty. The main idea is to select a short period of observation time to form a focused image, during which the motion of the target is approximately linear. Based on this principle, several ISAR-based autofocus algorithms are developed [20], [21]. However, ship motions are highly random, making it sometimes difficult to select an appropriate period. Besides, exploiting only part of the signal also causes the reduction of signal-to-noise ratio (SNR) and the loss of energy [22]. Another strategy is to correct range-variant phase errors in the range-Doppler domain, assuming that point scatterers in the same range gate have the same

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motion [23]. However, this may not be applicable for ships with azimuth-variant motions. By decomposing the image into subimages, some scholars realize refocusing of rotational ships via separately conducting phase error compensation for each subimage [24] or for each point scatterer [25]. Although these methods have relatively high accuracy, the main drawback of this strategy is the large computational load. In addition, neglecting the integration of the target may also cause geometric distortion. Therefore, it is still a challenge to refocus rotating ships effectively and efficiently in SAR imagery.

To address the above issues, this article proposes a new refocusing algorithm, which can improve the imaging quality of rotating ships in SAR imagery in a more effective and efficient way without energy loss. The nonlinear chirp scaling (NLCS) algorithm is originally proposed for squinted SAR data processing, during which the space variance of azimuth frequency modulation (FM) rate is balanced by multiplying a cubic phase perturbation term [26], [27], [28], [29]. Inspired by this, we attempt to apply the NLCS principle to correct space-variant phase errors for rotating ships by postprocessing single-look complex (SLC) SAR imagery. In the proposed algorithm, the effects of ship's rotations on SAR imaging are analyzed in detail and the mathematical model of phase errors is established, which provides the basis for the NLCS-based refocusing algorithm. Then, an equivalent azimuth inverse compression method is designed to recover the azimuth FM of the target image, during which the zero-padding operation can be avoided. After that, a modified NLCS operation taking cubic and quartic perturbation terms into consideration is derived to realize compensation for the space-variant phase errors. Specifically, the perturbation parameters are estimated adaptively via minimizing image entropy based on the constructed objective function. To the best of our knowledge, the proposed method is the first attempt to employ the NLCS principle to correct space-variant phase errors for the refocusing of rotating ships. In contrast to other refocusing techniques, there is no image decomposition or time window selection in the proposed method, and only two parameters are needed to be estimated in the NLCS operation for the whole image data. Therefore, the proposed method can achieve the refocusing of rotating ships more effectively while maintaining high efficiency and robustness.

The remainder of this article is organized as follows. Section II introduces the signal model of ship targets with rotational motions and the mathematical model of space-variant phase errors. Section III describes the proposed NLCS-based refocusing method and the estimation of perturbation parameters. In Section IV, simulated and real spaceborne SAR image data are processed to validate the effectiveness of the proposed method, and the conclusion is drawn in Section V.

II. Signal Model for Ship Targets

A. Geometry Model

Consider a spaceborne SAR system operating at side-looking mode, as shown in Fig. 1. The beam direction of the radar is approximately perpendicular to the flight direction of the satellite. Two coordinate systems are established. *O-XYZ* is the fixed space coordinate system, in which *X*-axis

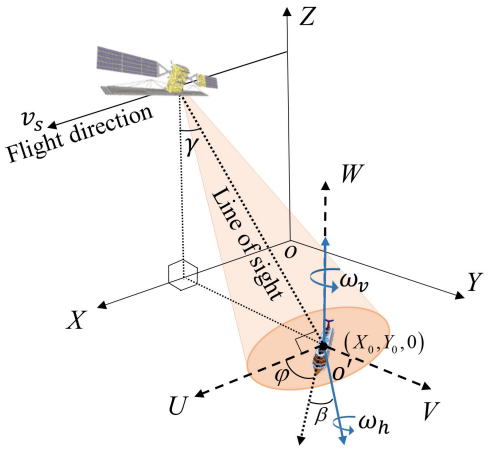


Fig. 1. Geometry of a spaceborne SAR system illuminating a rotating ship.

is parallel to the SAR platform flight direction. γ is the depression angle. *O′-UVW* is the fixed scene coordinate system obtained by translating the space coordinate system. The origin O' is located at the center of the scene to be imaged, whose coordinate in the *O-XYZ* is $(X_0, Y_0, 0)$.

Assume that a rotating ship is located at the center of the scene. φ is the angle between the U-axis and the direction of ship's bow. The rotations of a ship on the sea surface include roll, yaw, and pitch. Generally, roll and pitch are the most obvious motions [13]. The rotation of a ship target can be described by the horizontal rotation vector ω_h caused by roll and pitch, and the vertical rotation vector ω_v caused by yaw. For simplicity, we assume that the direction of the rotation vector is constant during the observation time. β is the angle between ω_h and the direction of the ship's bow. The instantaneous rotation angle at azimuth time t_a caused by the horizontal rotation vector is $\theta_h(t_a)$ and the rotation angle caused by the vertical rotation vector is $\theta_v(t_a)$. P is a point scatterer on the ship, whose position in the *O′-UVW* system is (u^p, v^p, w^p) when the ship is stationary. Considering ship rotational motions, the coordinate of P in the *O-XYZ* system at time t_a can then be expressed as

$$\begin{bmatrix} x_p(t_a) \\ y_p(t_a) \\ z_p(t_a) \end{bmatrix} = \begin{bmatrix} X_0 \\ Y_0 \\ 0 \end{bmatrix} + \mathbf{R}(\theta_v(t_a))\mathbf{R}(\beta+\varphi)\mathbf{R}_r[\theta_h(t_a)]\mathbf{R}(-\beta-\varphi)\begin{bmatrix} u^p \\ v^p \\ w^p \end{bmatrix} \tag{1}$$

where

$$\mathbf{R}(\cdot) = \begin{bmatrix} \cos(\cdot) & -\sin(\cdot) & 0 \\ \sin(\cdot) & \cos(\cdot) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

$$\mathbf{R}_r(\cdot) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\cdot) & -\sin(\cdot) \\ 0 & \sin(\cdot) & \cos(\cdot) \end{bmatrix}. \tag{3}$$

By making small-angle approximations to the trigonometric functions and ignoring the height of the target, the above

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expression can further be written as

$$\begin{bmatrix} x_p(t_a) \\ y_p(t_a) \\ z_p(t_a) \end{bmatrix} = \begin{bmatrix} X_0 + u^p \cos(\theta_v(t_a)) - v^p \sin(\theta_v(t_a)) \\ Y_0 + u^p \sin(\theta_v(t_a)) + v^p \cos(\theta_v(t_a)) \\ -u^p \theta_h(t_a) \sin(\beta+\varphi) + v^p \theta_h(t_a) \cos(\beta+\varphi) \end{bmatrix}. \tag{4}$$

The slant range error caused by target rotation can then be obtained by projecting position error vector along the radar line-of-sight (LOS) direction

$$\begin{aligned} \Delta R_p(t_a) = {} & u^p(\theta_h(t_a)\cos\gamma\sin(\beta+\varphi) - \theta_v(t_a)) \\ & - v^p \theta_h(t_a)\cos\gamma\cos(\beta+\varphi). \end{aligned} \tag{5}$$

The rotation of a ship target is an oscillatory angular motion, which can be modeled as sinusoidal functions [13]

$$\begin{cases} \theta_h(t_a) = A_h \sin\left(2\pi \dfrac{t_a}{T_h} + \varsigma_h\right) \\ \theta_v(t_a) = A_v \sin\left(2\pi \dfrac{t_a}{T_v} + \varsigma_v\right) \end{cases} \tag{6}$$

in which A_h and A_v represent the amplitude of rotational motion, T_h and T_v denote the oscillatory period, and ς_h and ς_v denote the initial phase. The rotation period of a typical destroyer is usually between 6.5 and 10.1 s [13]. The synthetic aperture time of a spaceborne SAR system can be approximately expressed as

$$T_a = \frac{\lambda R_0}{2\rho_a v_s} \tag{7}$$

in which λ is the wavelength, ρ_a is the azimuth resolution, v_s is the velocity of the SAR sensor platform, and R_0 represents the shortest slant range between the satellite and the target. For a nominal low Earth orbit (LEO) SAR system operating at X-band ($f_c = 10$ GHz) with 1-m resolution in azimuth, it is reasonable to consider that the synthetic aperture time is less than 3 s, smaller than the rotation period. Therefore, the variation of the rotation angle can be approximately modeled as a polynomial. Expanding (6) into its third-order Taylor series at $t_a = 0$ yields

$$\begin{cases} \theta_h(t_a) \approx k_0^h + k_1^h t_a + k_2^h t_a^2 + k_3^h t_a^3 \\ \theta_v(t_a) \approx k_0^v + k_1^v t_a + k_2^v t_a^2 + k_3^v t_a^3 \end{cases} \tag{8}$$

where

$$\begin{cases} k_0^h = A_h \sin\varsigma_h \\ k_1^h = A_h \dfrac{2\pi}{T_h}\cos\varsigma_h \\ k_2^h = -\dfrac{A_h}{2}\left(\dfrac{2\pi}{T_h}\right)^2 \sin\varsigma_h \\ k_3^h = -\dfrac{A_h}{6}\left(\dfrac{2\pi}{T_h}\right)^3 \cos\varsigma_h \end{cases} \quad \begin{cases} k_0^v = A_v \sin\varsigma_v \\ k_1^v = A_v \dfrac{2\pi}{T_v}\cos\varsigma_v \\ k_2^v = -\dfrac{A_v}{2}\left(\dfrac{2\pi}{T_v}\right)^2 \sin\varsigma_v \\ k_3^v = -\dfrac{A_v}{6}\left(\dfrac{2\pi}{T_v}\right)^3 \cos\varsigma_v. \end{cases} \tag{9}$$

Combining (5), (8), and (9), the slant range error caused by target rotation can further be written as

$$\begin{aligned} \Delta R_p(t_a) = {} & \left(u^p C_0 + v^p D_0\right) + \left(u^p C_1 + v^p D_1\right) t_a \\ & + \left(u^p C_2 + v^p D_2\right) t_a^2 + \left(u^p C_3 + v^p D_3\right) t_a^3 \end{aligned} \tag{10}$$

where

$$\begin{aligned} C_i &= \sin(\beta+\varphi)\cos\gamma k_i^h + k_i^v \\ D_i &= -\cos(\beta+\varphi)\cos\gamma k_i^h \quad i = 0, 1, 2, 3. \end{aligned} \tag{11}$$

Equation (10) shows that in addition to rotation parameters, the slant range error is also dependent on the position of the point scatterer.

B. Analysis of Phase Error

The proposed refocusing method for rotating ships is a postprocessing algorithm applied to SAR imagery where one or more moving ship targets may be present. In the SAR image domain, signals for moving targets exhibit additional phase errors in the azimuth frequency domain, while the range migration caused by target movement is generally considered to have been corrected during SAR imaging processing [2], [23], [30]. In this article, traditional chirp scaling algorithm (CSA) [31] is utilized to demonstrate the signal for rotating ships during SAR imaging processing. A typical CSA includes range cell migration correction (RCMC), range compression, and azimuth compression.

The instantaneous slant range between the spaceborne radar platform and the point scatterer P is given by

$$R_p(t_a) \approx R_{p,0} + \frac{v_s^2}{2R_{p,0}}\left(t_a - t^p\right)^2 + \Delta R_p\left(t_a - t^p\right) \tag{12}$$

where $R_{p,0}$ represents the shortest slant range between the satellite and the target and t^p denotes the time of the closest slant range for target P. The echo signal of target P can be expressed as [31]

$$\begin{aligned} S_p(t_r, t_a) = {} & \sigma_0 w_r\left(t_r - 2R_p(t_a)/c\right) w_a\left(t_a - t^p\right) \\ & \exp\left\{-j4\pi R_p(t_a)/\lambda\right\} \exp\left\{j\pi K_r\left(t_r - 2R_p(t_a)/c\right)^2\right\} \end{aligned} \tag{13}$$

where σ_0 denotes the backscatter amplitude, w_r is the range pulse envelope, w_a is the azimuth pulse envelop, t_r is the range time, K_r is the FM rate of the range pulse, and c is the speed of light.

After range compression and RCMC through the CSA, the signal of target P is aligned to the same range gate with phase errors caused by the target motion, which can be written as [30]

$$\begin{aligned} & S_{p,r}(t_r, t_a) \\ & \quad = \sigma_0 p_r\left(t_t - \frac{2R_{p,0}}{c}\right) w_a\left(t_a - t^p\right) \\ & \quad \exp\left\{-j\pi K_a\left(t_a - t^p\right)^2\right\} \exp\left\{-j4\pi \Delta R_p\left(t_a - t^p\right)/\lambda\right\} \end{aligned} \tag{14}$$

where p_r is the compressed pulse envelope and $K_a = 2v_s^2/\lambda R_{p,0}$ is the azimuth FM rate. The last term is the phase error caused by target rotation, which can be expressed as

$$\begin{aligned} & \frac{4\pi}{\lambda}\Delta R_p\left(t_a - t^p\right) \\ & \quad = \frac{4\pi}{\lambda}\left(u_p C_0 + v_p D_0\right) + \frac{4\pi}{\lambda}\left(u_p C_1 + v_p D_1\right)\left(t_a - t^p\right) \end{aligned}$$

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$$+\frac{4\pi}{\lambda}\left(u_pC_2+v_pD_2\right)\left(t_a-t^p\right)^2 +\frac{4\pi}{\lambda}\left(u_pC_3+v_pD_3\right)\left(t_a-t^p\right)^3. \tag{15}$$

The linear term in (15) results in a Doppler frequency shift, corresponding to an azimuthal shift in the final image. The quadratic and cubic terms, which are determined by rotation parameters and the target position, lead to a mismatch of the azimuth compression and thus can cause the azimuth defocusing of the SAR image [2], [23], [30], [32]. Since this article focuses on the refocusing of the target, the linear term is temporally ignored in the following analysis as it has no influence on image focus. Nevertheless, geometric distortion caused by linear term can still degrade the interpretability of the target, which may be discussed in future research.

Let v_g denote the ground velocity of the satellite. Then, the relationship between azimuth position u^p and azimuth position time t^p is given by

$$u^p=t^pv_g. \tag{16}$$

For reasons of simplicity, consider the simplified form for the range processed signal depending only on t_a. Assuming that point scatterer P is located at the lth range gate, the signal (14) can further be expressed as

$$\begin{aligned}S_{p,l,0}(t_a)=\sigma_0w_a\left(t_a-t^p\right)&\exp\left\{-j\pi K_a\left(t_a-t^p\right)^2\right\}\\&\exp\left\{-jt^p\left(B_2\left(t_a-t^p\right)^2+B_3\left(t_a-t^p\right)^3\right)\right\}\\&\exp\left\{j\left(A_{2,l}\left(t_a-t^p\right)^2+A_{3,l}\left(t_a-t^p\right)^3\right)\right\}\end{aligned} \tag{17}$$

$$\begin{cases}B_2=\dfrac{4\pi}{\lambda}v_gC_2\\ B_3=\dfrac{4\pi}{\lambda}v_gC_3\\ A_{2,l}=-\dfrac{4\pi}{\lambda}v_gD_3v^p\\ A_{3,l}=-\dfrac{4\pi}{\lambda}v_gD_3v^p.\end{cases} \tag{18}$$

Point scatterers located at the same range gate have the same range coordinate. Therefore, for the lth range gate, B_2, B_3, $A_{2,l}$, and $A_{3,l}$ are constants decided by radar wavelength, radar system pulse repetition frequency (PRF), and rotation parameters. In (17), the first phase term is introduced by the motion of the satellite platform, which contributes to SAR image formation. The second and third phase terms are caused by target rotation. The second term varies with the scatterer's azimuth position, while the third term depends on the range gate.

The azimuth compression is usually conducted in the azimuth frequency domain by a matched filter whose parameter is updated for each range gate. The matched filtering function is given by [31]

$$H_1(f_a)=\exp\left\{-j\pi\frac{f_a^2}{K_a}\right\} \tag{19}$$

where f_a represents the azimuth frequency. In order to analyze the effect of phase errors on azimuth compression, (17) is transformed to azimuth frequency domain according to the principle of stationary phase (POSP) and multiplied by the matched filtering function given in (19). Ignoring the envelope and constant terms, the result can be approximately expressed as

$$\begin{aligned}S_{p,l,1}(f_a)&\approx\exp\left\{-j2\pi f_at^p\right\}\exp\left\{j\left(-\frac{t^pB_2f_a^2}{K_a^2}+\frac{t^pB_3f_a^3}{K_a^3}\right)\right\}\\&\exp\left\{j\left(\frac{A_{2,l}f_a^2}{K_a^2}-\frac{A_{3,l}f_a^3}{K_a^3}\right)\right\}\exp\{j\zeta(f_a)\}.\end{aligned} \tag{20}$$

Note that the last term is induced by ship velocity along the azimuth direction, which is temporarily neglected in the previous analysis for simplicity. The azimuth compressed signal for target P in the final SAR image can be obtained by transforming (20) to azimuth time domain via azimuth inverse fast Fourier transform (IFFT).

For a stationary target, the signal after matched filtering only consists of constant and linear phase terms in the azimuth frequency domain. However, for a rotating target, the residual high-order phase terms, which cannot be eliminated via classical azimuth compression processing, will cause azimuth defocusing and image quality degradation. In SAR images, ship targets, such as destroyers and aircraft carriers, are usually composed of multiple point scatterers. Formula (20) shows that phase errors caused by rotations depend on the azimuth and range position of these point scatterers. The 2-D space-variant phase errors cause the space variance of defocusing, which will be addressed in Section III.

III. Refocusing of Ship Target

Having analyzed the impacts of ship rotation on SAR imaging, this section is set to give a refocusing algorithm to correct phase errors of ship targets in SAR imagery. Taking both translational and rotational motions into consideration, phase errors of signals for moving ships can be divided into three types. The first one is the space-invariant phase error identical for all point scatterers. The second one is the range-variant azimuth-invariant phase error, corresponding to the third term in (20). This phase error varies slowly with slant range but is identical for scatters located at the same range gate. The third one is the azimuth-variant phase error induced by ship rotation, corresponding to the second term in (20). This phase error varies with the azimuth position of the point scatterer. The first and the second type of phase errors can be solved by conducting a 2-D phase compensation in the range-Doppler domain via classical autofocusing algorithm. The azimuth-variant phase errors, which are unable to be corrected simultaneously in azimuth frequency domain, will be preciously corrected by the proposed NLCS-based method.

The novel refocusing scheme is as follows. After extracting a ship target from the SAR image, traditional metric-based refocusing method is employed for azimuth-invariant phase error compensation. Next, azimuth inverse compression is operated to make the data suitable for next processing. Then, azimuth-variant phase errors are corrected by the NLCS-based algorithm via multiplying a high-order perturbation function and the refocused target image can be finally obtained.

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A. Azimuth-Invariant Phase Error Compensation

Through azimuth FFT, the extracted ship target image is transformed into range-Doppler domain, in which the azimuth signal is shown as (20). The phase errors induced by the last two terms do not change with the target azimuth position, so they can be corrected in the range-Doppler domain by the traditional refocusing method. In this article, the contrast-based algorithm [7] is implemented to correct azimuth-invariant phase errors. The range variance of phase error is solved by dividing the data into several blocks along the range. Because of the existence of residual azimuth-variant phase errors, it is not possible to make all point scatterers focused well at this step. Point scatterers with stronger energy tend to contribute more to the image contrast. Therefore, after contrast-based autofocusing processing, scatterer with strongest energy is supposed to be corrected well and focused best. According to (20), the scatterer located at the rotation center has no phase error induced by ship rotation. In order to match the signal with the mathematical model given in (20), the well-focused strongest point scatterer is shifted to the image center along the azimuth direction for each range block. To avoid data overflow, the azimuth size of the extracted image should be approximately twice the actual azimuth size occupied by the target. In addition, when extracting the target image, it should be ensured that the target is located at the center of the image.

Now, the azimuth-invariant phase errors are corrected and the rotation center of the ship target is located at $t_a = 0$. For the sake of convenience, it is assumed that the strongest point after azimuth-invariant phase error correction is just located at the azimuth center. The signal in the azimuth frequency domain can then be expressed as

$$S_{p,2}(f_a) \approx \exp\left\{ j\left(-B_2 t^p \frac{f_a^2}{K_a^2} + B_3 t^p \frac{f_a^3}{K_a^3} - 2\pi f_a t^p \right) \right\}. \tag{21}$$

Compared with (20), the subscript l corresponding to range gate is eliminated in (21) because the range-variant phase errors have been corrected. This means that signals of all range gates can be processed together in the next processing.

B. Azimuth Inverse Compression

Equation (21) shows that it is difficult to correct residual phase errors of all point scatterers directly in the azimuth frequency domain as signals overlap with each other. This problem is solved by the NLCS-based refocusing method in this article. A classical NLCS algorithm is generally embedded before azimuth compression in imaging processing algorithms and cannot directly be used to process image data. Therefore, image data should be transformed into an azimuth uncompressed domain to recover the azimuth FM characteristics. This can be achieved by multiplying the conjugate of matched filtering function given by (19) in the azimuth frequency domain [34]. To avoid azimuth aliasing, zero-padding operation is needed before inversing. Generally, the number of azimuth pixels after zero padding is supposed to be equal or larger than that of raw data. For a typical SAR system operating with 1-m resolution in azimuth, a destroyer or an aircraft carrier usually occupies approximately 100–200 pixels along azimuth in the SAR image, while its raw signal has thousands of sampling points. Zero-padding operation may lead to a significant increasement in computational load. Therefore, an equivalent azimuth inverse compression without zero-padding operation is proposed to improve the processing efficiency.

TABLE I
PARAMETERS OF THE SPACEBORNE SAR SYSTEM

Parameters	Value
Radar Wavelength	0.03125m
PRF	8000Hz
Obit Height	600Km
Bandwidth	150MHz
Depression Angle	25°
Sampling Rate	180MHz

Let N denote the number of azimuth pixels in the extracted image. The inverse compression function is given by

$$H_2(f_a) = \exp\left\{ j\pi \frac{f_a^2}{K_e} \right\} \tag{22}$$

where K_e represents the FM rate used during the inversing operation. Multiplying (21) and (22) yields

$$S_{p,3}(f_a) \approx \exp\left\{ j\left(-\frac{B_2' t^p f_a^2}{K_e^2} + \frac{B_3' t^p f_a^3}{K_e^3} - 2\pi f_a t^p + \frac{\pi f_a^2}{K_e} \right) \right\} \tag{23}$$

where $B_2' = B_2(K_e^2/K_a^2)$ and $B_3' = B_3(K_e^3/K_a^3)$. Transforming (23) to azimuth time domain by azimuth IFFT achieves

$$\begin{aligned} S_{p,4}(t_a) = {} & w_a\left(t_a - t^p\right) \\ & \exp\left\{ -j\pi K_e \left(t_a - t^p\right)^2 \right\} \\ & \exp\left\{ -j\left(B_2' t^p \left(t_a - t^p\right)^2 + B_3' t^p \left(t_a - t^p\right)^3 \right) \right\}. \end{aligned} \tag{24}$$

The number of pixels occupied by the pulsewidth of a point target signal given by $S_{p,4}(t_a)$ is about f_{prf}^2 / K_e, where f_{prf} is the system PRF. In order to avoid azimuth time-domain aliasing and ensure the effectiveness of the POSP, the following condition should be satisfied:

$$2f_{prf}^2/N > K_e > f_{prf}^2/N. \tag{25}$$

The purpose of this step is to recover the azimuth FM of the target image and to make sure that the space-variant phase errors can be balanced by perturbation terms. The size of data to be processed can be dramatically decreased by the proposed equivalent azimuth inverse compression method.

C. Azimuth-Variant Phase Error Compensation Via NLCS

A conventional NLCS method is designed to compensate for the linear variation of azimuth FM rate during SAR imaging processing. However, for signals of rotating ships, the space variance of phase errors is more complex and the compensation precision of traditional NLCS method is

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TABLE II
IMAGE QUALITY MEASUREMENT RESULTS IN AZIMUTH

Point Scatters		PGA Method [3]	ISAR-based Method [19]	Proposed Method	
				With cubic perturbation term	With cubic and quartic perturbation terms
P_1	IRW(m)	\	2.21	1.31	**1.29**
	PSLR(dB)	\	-4.40	-10.03	**-12.21**
P_2	IRW(m)	1.72	2.07	1.38	**1.37**
	PSLR(dB)	-5.21	-9.51	**-12.72**	**-12.72**
P_4	IRW(m)	1.54	2.08	**1.45**	**1.45**
	PSLR(dB)	-6.95	-10.56	-12.14	**-12.82**
P_5	IRW(m)	\	2.17	1.48	**1.46**
	PSLR(dB)	\	-6.03	-10.01	**-12.38**

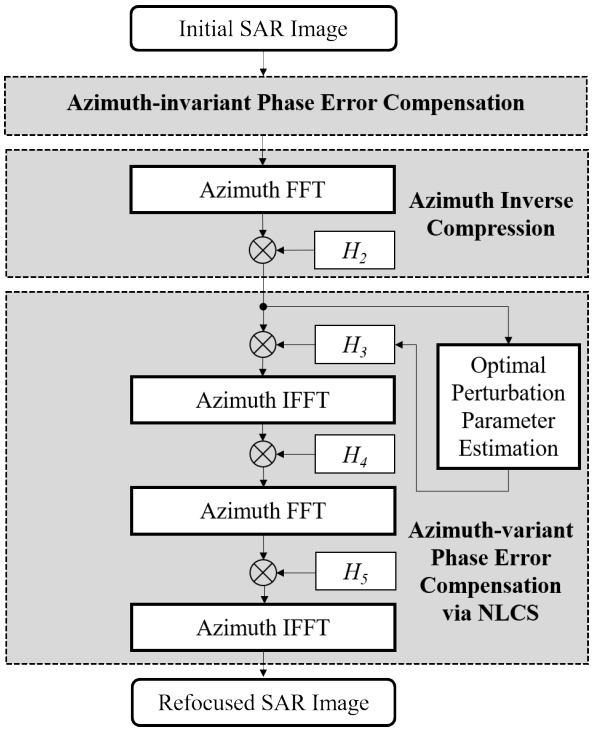


Fig. 2. Flowchart of the proposed refocusing algorithm for rotating ships.

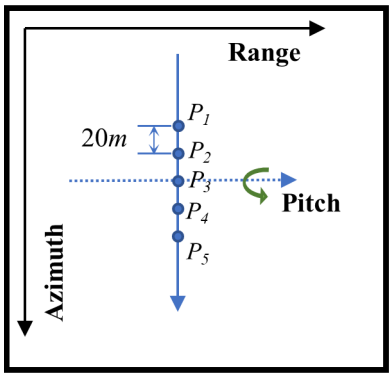


Fig. 3. Layout of the imaging scene for the first simulation.

insufficient. In order to improve the compensation accuracy, a modified NLCS operation taking higher order perturbation terms into consideration is proposed.

TABLE III
MOTION PARAMETERS OF THE SHIP

Case number	Translational motion		Rotational motion		
	Velocity (m/s)	Direction (°)	Amplitude (°)	Period (s)	Dimensionality
1	0	145	0	/	Roll
			0.9	9	Pitch
			0	/	Yaw
2	0	100	5	15	Roll
			0	/	Pitch
			0	/	Yaw
3	0	60	0	/	Roll
			0	/	Pitch
			1.5	10	Yaw
4	10	120	5	15	Roll
			0.9	9	Pitch
			1.5	10	Yaw

TABLE IV
PARAMETERS OF SPACEBORNE SAR SYSTEMS

Satellite	Parameters	Value
PJ-2	Radar Wavelength	0.03123m
	PRF	4900Hz
	Obit Height	600Km
	Bandwidth	270MHz
GF-3	Radar Wavelength	0.05552m
	PRF	3682Hz
	Obit Height	750Km
	Bandwidth	240MHz
ICEYE	Radar Wavelength	0.03109m
	PRF	5301Hz
	Obit Height	569Km
	Bandwidth	300MHz

In (24), the phase error needed to be compensated for is

$$\Theta = B_2' t^p \left(t_a - t^p\right)^2 + B_3' t^p \left(t_a - t^p\right)^3 \tag{26}$$

which consists of the quadratic and cubic terms of azimuth time t_a. The second-order and the third-order coefficients vary linearly with azimuth position time t^p. Based on the principle of the NLCS, the azimuth-variant phase errors can be removed by a perturbation function. In the proposed algorithm, cubic and quartic perturbation terms are considered. The perturbation function is modeled as

$$H_3(f_a) = \exp\left\{ j\alpha f_a^3 + j\xi f_a^4 \right\} \tag{27}$$

where α and ξ are perturbation parameters needed to be determined. Multiplying (23) and (27) and transforming the

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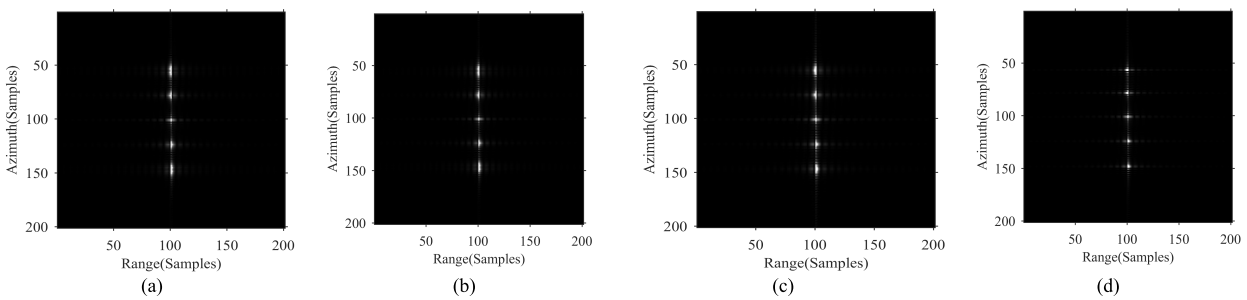


Fig. 4. SAR images of the five point scatterers. (a) Result after SAR imaging processing. (b) Result refocused by the PGA method. (c) Result refocused by the ISAR-based method. (d) Result refocused by the proposed method.

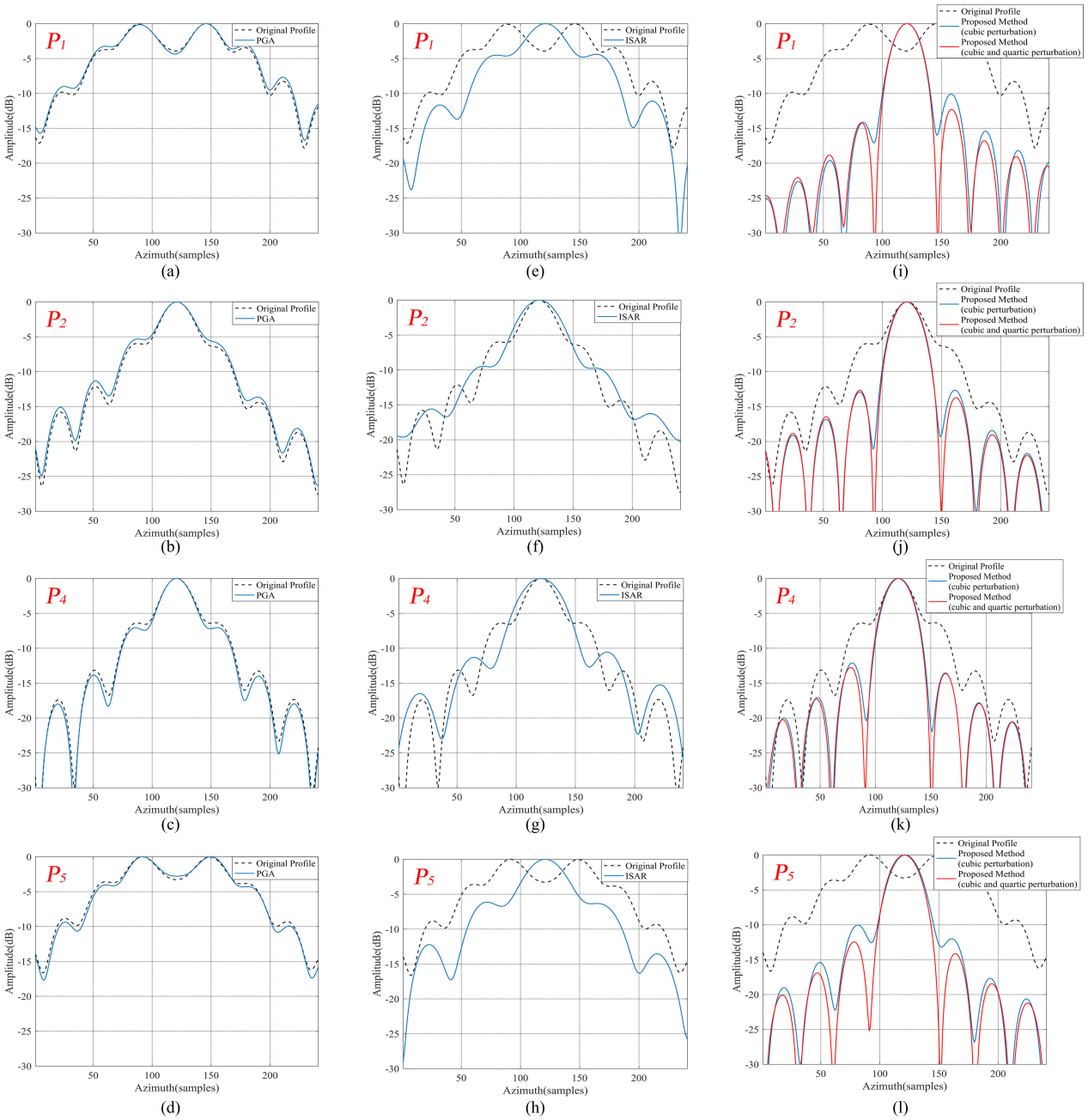


Fig. 5. Azimuth profiles of point scatterers P_1, P_2, P_4, and P_5 achieved by different methods. (a)–(d) Results by the PGA method. (e)–(h) Results by the ISAR-based method. (i)–(l) Results by the proposed refocusing method.

result to azimuth time domain via azimuth IFFT yields

$$S_{p,5}(t_a) = w_a\left(t_a - t^p\right) \exp\left\{-j\pi K_e\left(t_a - t^p\right)^2\right\} \exp\left\{-jB_2' t^p\left(t_a - t^p\right)^2\right\} \exp\left\{-jB_3' t^p\left(t_a - t^p\right)^3\right\}$$

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TABLE V
DETAILS OF THE DATA SOURCES

Sensor	PJ-2				GF-3			ICEYE	
Ship Number	1	2	3	4	1	2	3	1	2
Operation Mode	Slide-Spotlight				Spotlight			Spotlight	
Resolution	1m				1m			1m	
Date of Data Acquisition	20220507		20190416		20200912	20201114		20210406	
Illumination Area	Strait of Gibraltar		Port of Qingdao		Port of Dalian	Port of Fuzhou		Port of Rotterdam	

TABLE VI
NUMERICAL RESULTS OF SAR IMAGES PROCESSED BY DIFFERENT METHODS

Sensor		PJ-2				GF-3			ICEYE	
Ship Number Image Size (pixel)		1 226× 238	2 396× 304	3 266× 329	4 400× 487	1 219× 228	2 255× 171	3 247× 101	1 259× 228	2 340× 119
Image Entropy	Raw Image	7.525	8.048	6.954	8.423	7.301	7.837	7.540	8.520	8.228
	PGA Method[3]	7.224	8.067	6.574	8.325	7.234	7.771	7.148	8.229	7.705
	ISAR-based Method[19]	7.410	7.956	6.662	8.412	7.205	7.788	6.900	8.235	7.231
	Proposed Method	**6.974**	**7.575**	**6.518**	**8.138**	**7.012**	**7.629**	**6.603**	**8.058**	**7.108**
Image Contrast	Raw Image	7.289	11.126	14.128	9.732	5.959	7.101	3.737	2.773	3.572
	PGA Method[3]	7.790	10.986	14.528	9.611	5.954	7.180	3.810	2.868	3.939
	ISAR-based Method[19]	7.820	11.332	15.308	9.772	6.184	7.269	5.693	2.950	**5.099**
	Proposed Method	**9.231**	**13.421**	**15.778**	**11.118**	**6.354**	**7.808**	**7.121**	**2.999**	4.472
Peak Power	Raw Image	555.2	254.74	911.216	205.24	52554	20159	7864	3457	2232
	PGA Method[3]	670.3	234.268	1393.85	277.34	71021	26371	**17390**	5676	4089
	ISAR-based Method[19]	506.6	244.98	1218.5	207.54	50962	21879	4174	3237	2936
	Proposed Method	**689.0**	**373.034**	**1394.57**	**279.40**	**72766**	**26400**	11325	**6362**	**6297**
Cost(s)	PGA Method[3]	0.111	0.216	0.177	0.384	0.115	0.097	0.43	0.141	0.135
	ISAR-based Method[19]	0.617	1.152	0.611	2.108	0.557	0.459	0.384	0.708	0.847
	Proposed Method	0.520	0.751	0.461	1.118	0.389	0.422	0.381	0.551	0.454

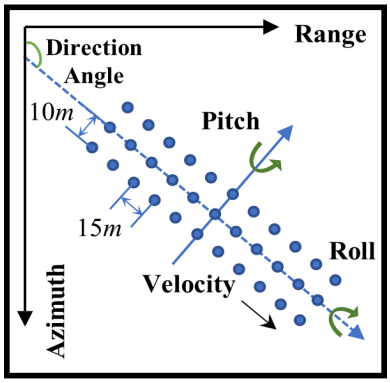


Fig. 6. Layout of the imaging scene for the second simulation.

$$\begin{aligned}&\exp\left\{-j\alpha K_e^3\left(t_a-t^p\right)^3+j\xi K_e^4\left(t_a-t^p\right)^4\right\}\\&=w_a\left(t_a-t^P\right)\\&\quad\exp\left\{j\left(\Phi_0+\Phi_1 t_a+\Phi_2 t_a^2+\Phi_3 t_a^3+\Phi_4 t_a^4\right)\right\}\end{aligned}\tag{28}$$

where

$$\begin{cases}\Phi_1=2\pi K_e t^p+2B_2'\left(t^p\right)^2-3B_3'\left(t^p\right)^3\\-3\alpha K_e^3\left(t^p\right)^2-4\xi K_e^4\left(t^p\right)^3\\\Phi_2=-\pi K_e-B_2't^p+3\left(t^p\right)^2B_3'\\+3t^p\alpha K_e^3+6\left(t^p\right)^2\xi K_e^4\\\Phi_3=-B_3't^p-\alpha K_e^3-4t^p\xi K_e^4\\\Phi_4=\xi K_e^4.\end{cases}\tag{29}$$

For simplicity, the cubic and quantic phases are ignored when calculating stationary phase point in the previous azimuth FFT and azimuth IFFT operations as their values are too small to affect the result [31], [33]. In order to eliminate the space-variant phase errors, the following equations should be satisfied:

$$\begin{cases}-B_2't^p+3t^p\alpha K_e^3=0\\-B_3't^p-4t^p\xi K_e^4=0.\end{cases}\tag{30}$$

Solving the above equations, the undetermined perturbation parameters can be obtained as follows:

$$\begin{cases}\alpha=\dfrac{B_2'}{3K_e^3}\\\xi=-\dfrac{B_3'}{4K_e^4}.\end{cases}\tag{31}$$

The compensation filter in the azimuth time domain can then be expressed as

$$H_4(t_a)=\exp\left\{-j\xi K_e^4t_a^4+j\alpha K_e^3t_a^3\right\}.\tag{32}$$

After multiplying (28) and (32), high-order terms of t_a (i.e., $\Phi_3t_a^3$ and $\Phi_4t_a^4$) are removed. The coefficients of the first and second terms of t_a are given by

$$\begin{cases}\Phi_1'=2\pi K_et^p+B_2'\left(t^p\right)^2-2B_3'\left(t^p\right)^3\\\Phi_2'=-\pi K_e+1.5B_3'\left(t^p\right)^2.\end{cases}\tag{33}$$

Since the ground velocity v_g is about several thousands of meters per second and the length of a ship is generally no more than 350 m, the variation range of t^p is much less than 0.1 s. Therefore, the quadratic and higher order terms of t^p are neglected and the signal can be approximately written as

$$S_{p,6}(t_a)=w_a\left(t_a-t^p\right)\exp\left\{-j\pi K_e\left(t_a-t^p\right)^2\right\}.\tag{34}$$

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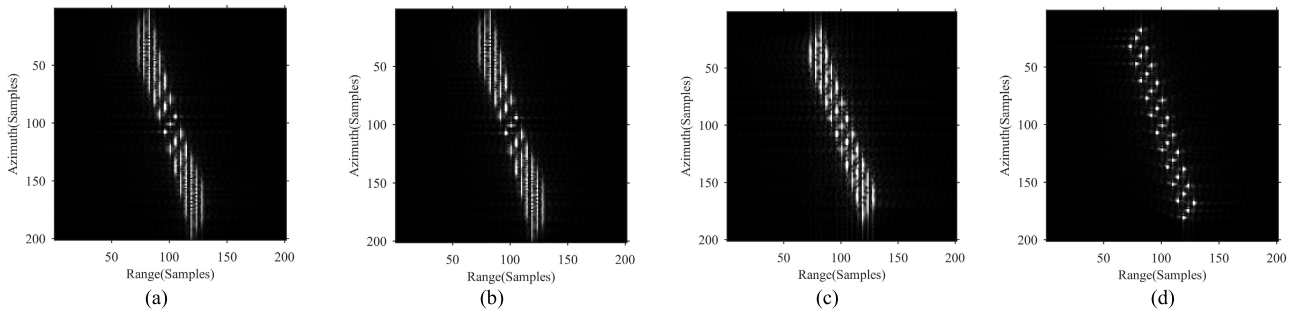


Fig. 7. Images of the ship target for case 1. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

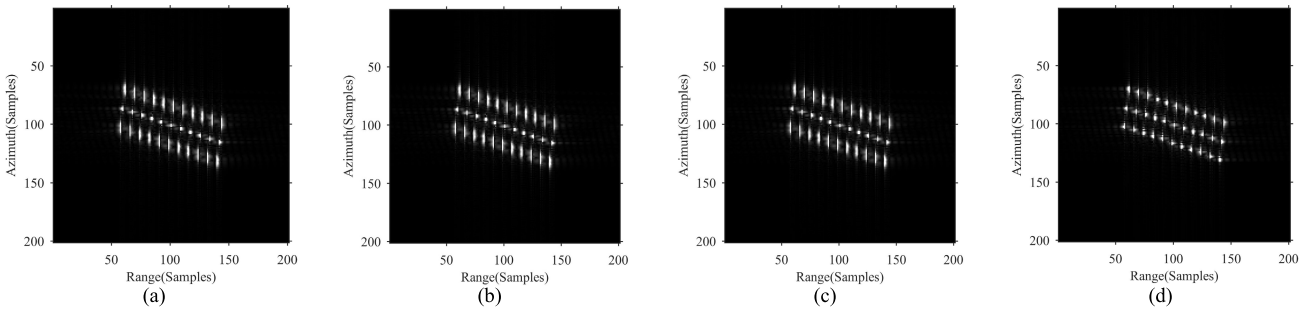


Fig. 8. Images of the ship target for case 2. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

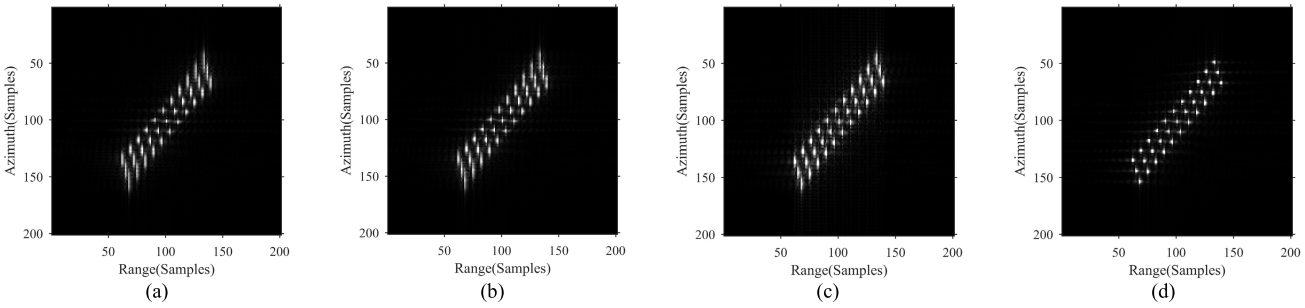


Fig. 9. Images of the ship target for case 3. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

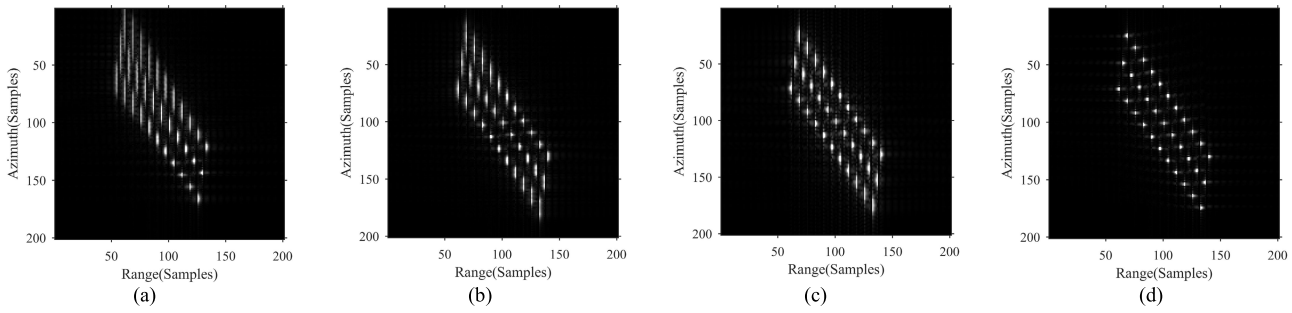


Fig. 10. Images of the ship target for case 4. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

Compared with (24), the space variance of phase errors has been eliminated in (34). All point scatterers share the same equivalent FM rate K_e, which can be corrected in the azimuth frequency domain by the following matched filter:

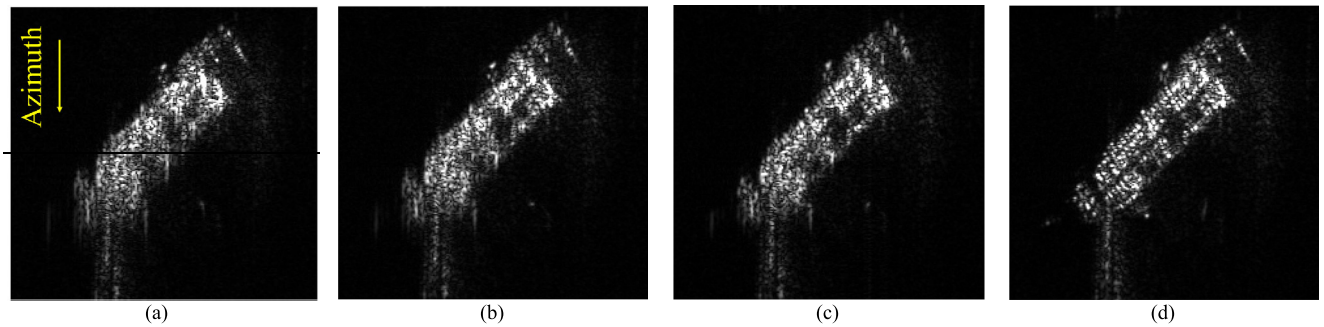
$$H_5(f_a) = \exp\left\{-j\pi \frac{f_a^2}{K_e}\right\}. \tag{35}$$

The final refocused image can be acquired by transforming the signal back to azimuth time domain.

D. Perturbation Parameter Estimation

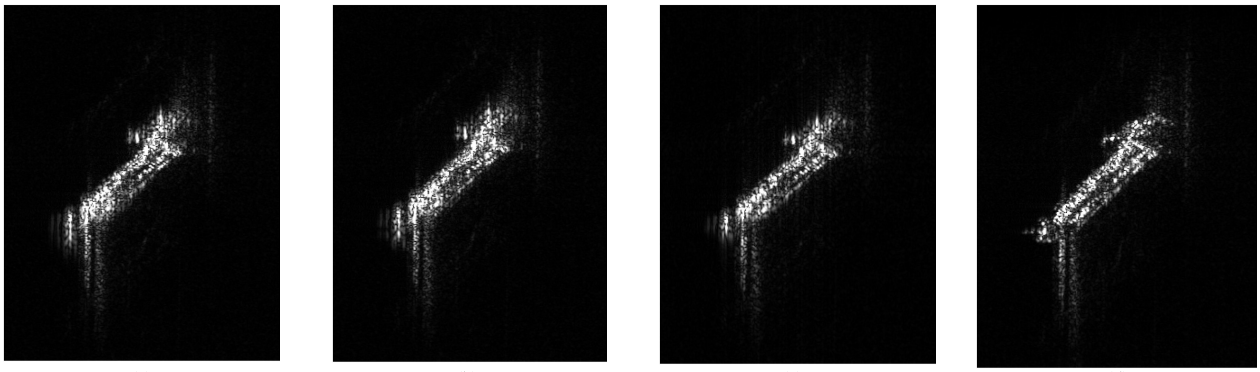
Equation (31) shows that the perturbation parameters are determined by rotation parameters of the ship target. However, motion information is unpredictable in practice. Note that

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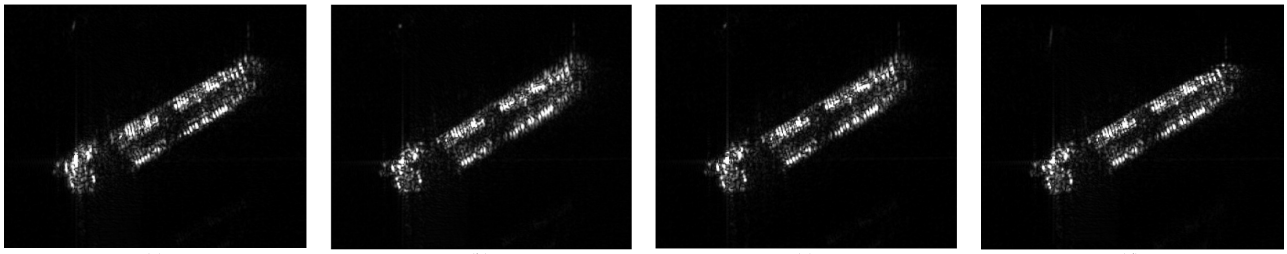
(a) (b) (c) (d)

Fig. 11. Images of the first ship target of PJ-2. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.



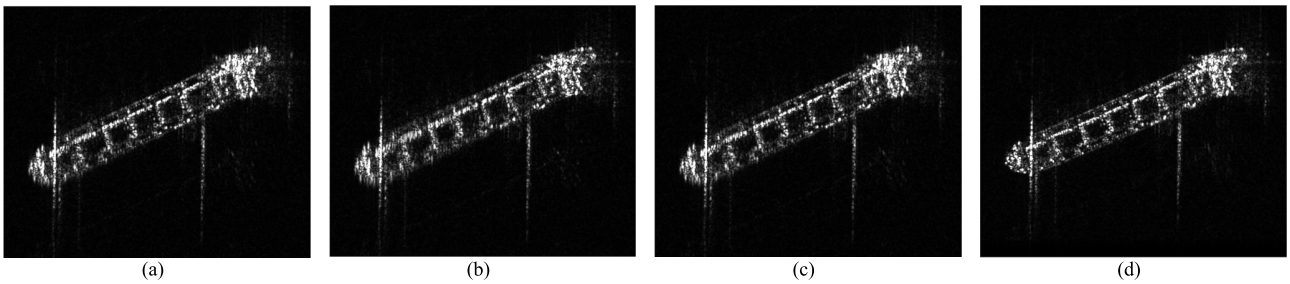
(a) (b) (c) (d)

Fig. 12. Images of the second ship target of PJ-2. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.



(a) (b) (c) (d)

Fig. 13. Images of the third ship target of PJ-2. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.



(a) (b) (c) (d)

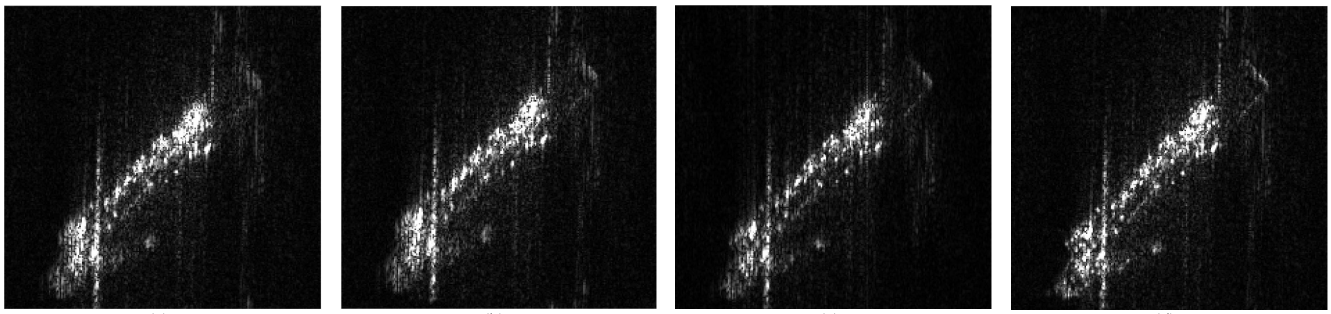
Fig. 14. Images of the fourth ship target of PJ-2. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

phase errors cause the degradation of image quality. Therefore, perturbation parameters can be estimated via optimizing image quality. Let $I(l, m; \tilde{\alpha}, \tilde{\xi})$ be the complex-valued image of the ship target processed by a perturbation function with $(\tilde{\alpha}, \tilde{\xi})$, where l denotes range pixel index and m denotes azimuth pixel index. The image entropy is defined as follows [32]:

$$\begin{aligned} IE(\tilde{\alpha}, \tilde{\xi}) &= -\sum_{l=0}^{L-1}\sum_{m=0}^{M-1} \frac{\left|I\left(l, m; \tilde{\alpha}, \tilde{\xi}\right)\right|^2}{Q} \ln \frac{\left|I\left(l, m; \tilde{\alpha}, \tilde{\xi}\right)\right|^2}{Q} \\ &= -\frac{1}{Q}\sum_{l=0}^{L-1}\sum_{m=0}^{M-1} \left|I\left(l, m; \tilde{\alpha}, \tilde{\xi}\right)\right|^2 \ln\left|I\left(l, m; \tilde{\alpha}, \tilde{\xi}\right)\right|^2 + \ln Q \end{aligned} \tag{36}$$

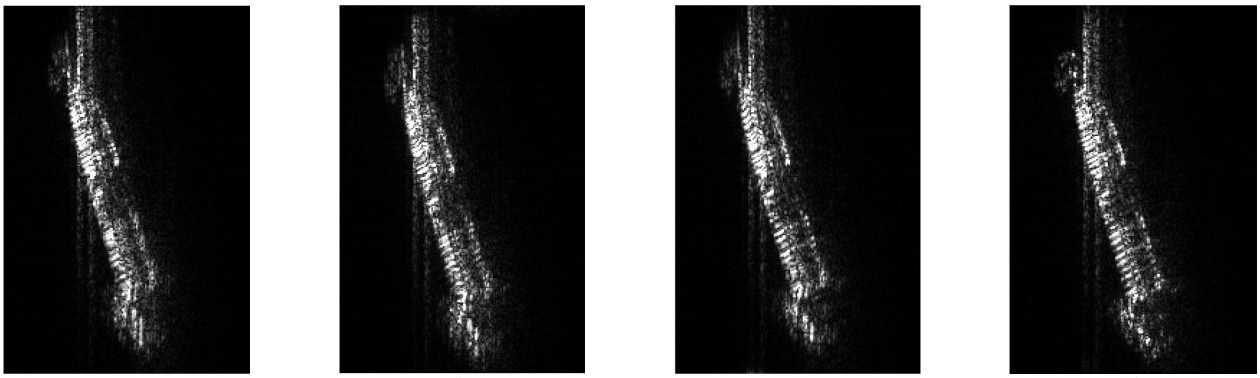
where $Q = \sum_{l=0}^{L-1}\sum_{m=0}^{M-1} |I(l, m; \tilde{\alpha}, \tilde{\xi})|^2$ represents the energy of the signal and is a constant. Image entropy gives a measure

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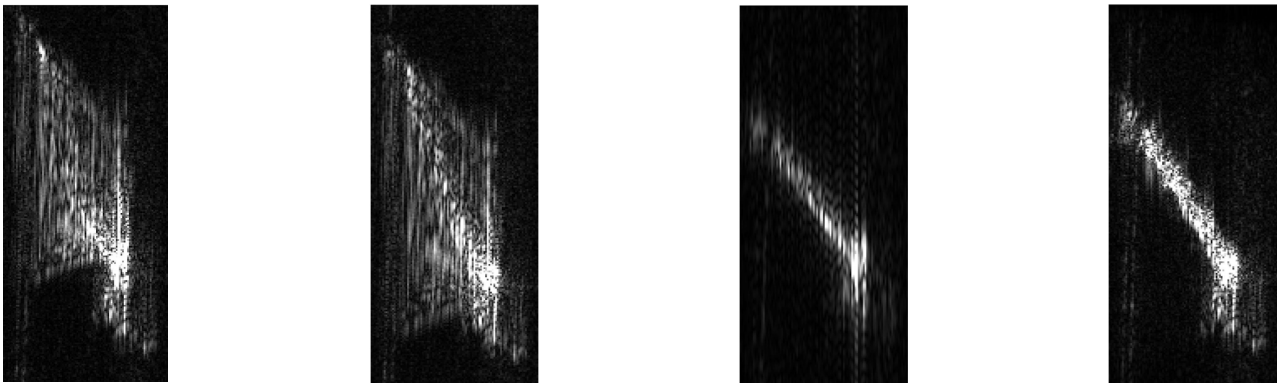
(a) (b) (c) (d)

Fig. 15. Images of the first ship target of GF-3. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.



(a) (b) (c) (d)

Fig. 16. Images of the second ship target of GF-3. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.



(a) (b) (c) (d)

Fig. 17. Images of the third ship target of GF-3. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

of image focusing, with lower entropy indicating higher image quality. Therefore, the estimation of perturbation parameters can be achieved by minimizing the image entropy. The objective function to be solved is constructed as

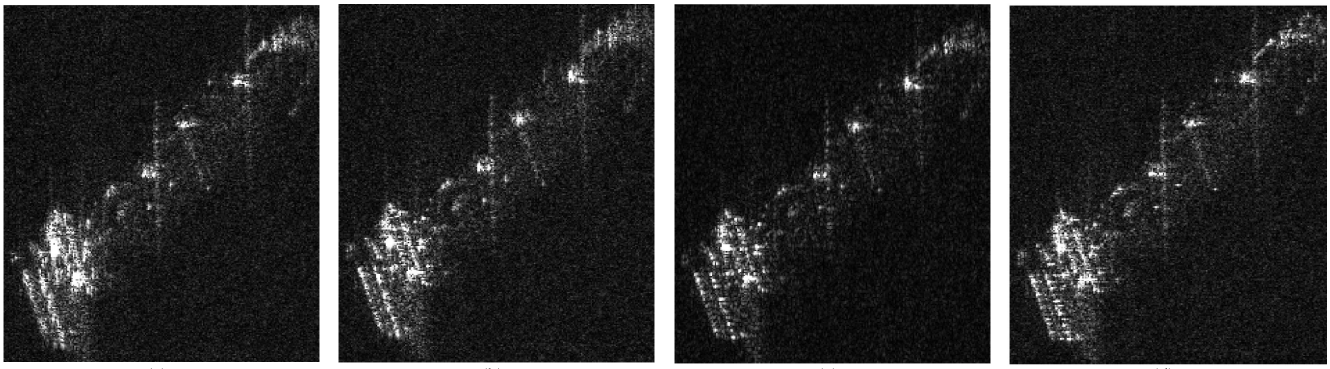
$$\left(\hat{\alpha},\hat{\xi}\right)=\arg\left(\min_{\tilde{\alpha},\tilde{\xi}}\left[IE\left(\tilde{\alpha},\tilde{\xi}\right)\right]\right). \tag{37}$$

Since there are only two parameters to be estimated, peak search is utilized to find $\hat{\alpha}$ and $\hat{\xi}$. Combining (9), (11), (18), and (31), the maximum absolute values of α and ξ, which determine the search range, are given by

$$\begin{cases} |\alpha|_{\max} < \dfrac{(2\pi)^3 v_g}{3\lambda K_a^2 K_e}\left(\dfrac{|A_h|_{\max}}{|T_h|^2_{\max}}+\dfrac{|A_v|_{\max}}{|T_v|^2_{\max}}\right) \\ |\xi|_{\max} < \dfrac{(2\pi)^4 v_g}{12\lambda K_a^3 K_e}\left(\dfrac{|A_h|_{\max}}{|T_h|^3_{\max}}+\dfrac{|A_v|_{\max}}{|T_v|^3_{\max}}\right). \end{cases} \tag{38}$$

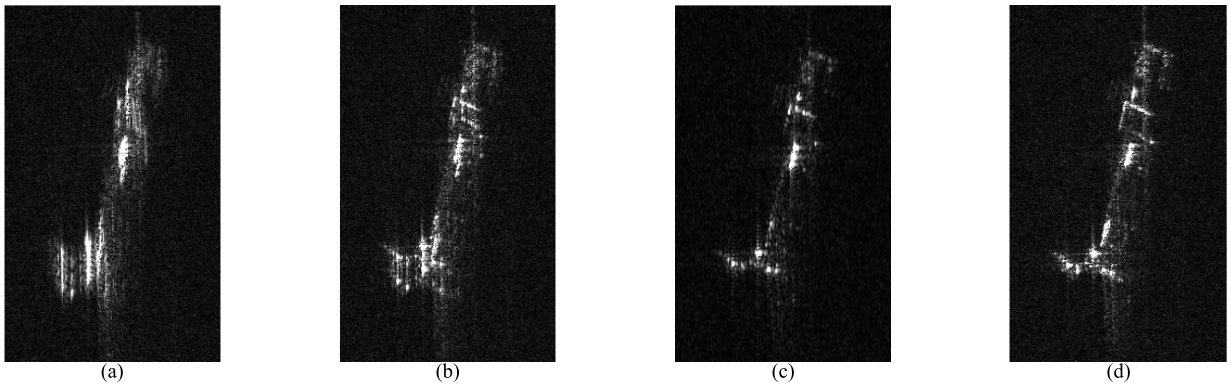
The optimal values of α and ξ can be obtained by computing image entropy for a set of $\left(\tilde{\alpha},\tilde{\xi}\right)$ values and finding the distribution peak by a 2-D search. Another more efficient scheme is to decompose the 2-D search into two 1-D searches. First, let $\tilde{\xi}=0$ and get the optimal estimate $\hat{\alpha}$ of α. Then, set $\tilde{\alpha}=\hat{\alpha}$ and find the optimal estimate $\hat{\xi}$ of ξ. It is worth noting that only

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(a) (b) (c) (d)

Fig. 18. Images of the first ship target of ICEYE. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.



(a) (b) (c) (d)

Fig. 19. Images of the second ship target of ICEYE. (a) Image after SAR imaging processing. (b) Image refocused by the PGA method. (c) Image refocused by the ISAR-based method. (d) Image refocused by the proposed NLCS-based method.

two parameters are needed to be estimated for the whole image data during azimuth-variant phase error correction, making it more efficient than other methods that generally require much more parameters. After azimuth-variant phase error correction, data of each range block should be shifted back to the original position. This step can be achieved by multiplying with a linear phase term in the azimuth frequency domain in conjunction with azimuth compression operation.

Fig. 2 shows the flowchart of the proposed refocusing algorithm, with three main steps: azimuth-invariant phase error compensation, azimuth inverse compression, and azimuth-variant phase error compensation based on the NLCS principle. In the third step, two perturbation parameters are needed to be estimated by optimizing image entropy.

IV. Experimental Result

A. Simulated Data

In this section, simulations are performed to demonstrate the effectiveness of the refocusing algorithm. Spaceborne SAR signals are generated using parameters shown in Table I.

In the first simulation, five point scatterers located in the same range gate are constructed, denoting the perpendicular of a ship. The distance between the adjacent point scatterers is 20 m. A pitch with an amplitude of 0.5° and a period of 9 s is added to the point targets, as shown in Fig. 3. The SAR image after imaging processing via CSA [31] is shown in Fig. 4(a), in which target P_3 is well focused, while P_1, P_2, P_4, and P_5 experience obviously defocusing because of the pitch. Besides, as the increase of the distance between the point target and the rotation center P_3, the degree of defocusing becomes more severe. P_1 and P_5 suffer from worse defocusing compared with P_2 and P_4. Then, the proposed NLCS-based refocusing method is utilized to process the image data. The PGA method in [3] and the ISAR-based method in [19] are selected as the reference methods to make a comparison. The results by the PGA and ISAR-based methods are shown in Fig. 4(b) and (c). The result by the proposed method is shown in Fig. 4(d). From a visual point of view, only Fig. 4(d) is well-focused, in which all point scatterers are clear and visible. Point scatterers in Fig. 4(b) are still as smeared as those in the original image [i.e., Fig. 4(a)]. The degree of focusing in Fig. 4(c) is improved compared to the original result. However, there is still some slight defocusing.

To highlight the differences between the results, the azimuth profiles of point scatterers P_1, P_2, P_4, and P_5 processed by various methods are extracted and illustrated in Fig. 5. Black dashed lines indicate the original profiles of point scatterers, while the solid lines represent the profiles after refocusing processing. In addition, the azimuth impulse response width (IRW) and peak sidelobe ratio (PSLR) of these point scatterers are measured for a qualitative evaluation. The results are listed in Table II. The point scatterers processed by the proposed method are much better focused than those processed by the PGA and ISAR-based methods, as indicated by the improved IRW values and reduced PSLR values. Besides, Fig. 5(i)–(l) shows the results by the proposed method with different perturbation terms. Results refocused using the cubic and

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quartic perturbation terms have better IRW and PSLR values compared to those refocused using only the cubic perturbation term, especially for point scatterers at the azimuth edge. This suggests that the refocusing accuracy of the proposed method improves as the order of the perturbation term increases. It should be noted that the IRW values of each point scatterer after refocusing are different. This is because motion errors induced by target rotations vary among point scatterers, leading to different changes in Doppler bandwidth.

In the second simulation, 33 point scatterers are constructed to indicate a ship target with a size of 20 × 150 m. The layout of the imaging scene is shown in Fig. 6. Table III gives the motion parameters of the ship target under different cases, including roll, pitch, yaw, and sailing velocity. Figs. 7(a)–10(a) show the SAR images of the ship target after imaging processing. There exists obvious space-variant defocusing in these images. The processing results by the PGA method are shown in Figs. 7(b)–10(b), and the results by the ISAR-based method are shown in Figs. 7(c)–10(c). Figs. 7(d)–10(d) show the refocusing results by the proposed NLCS-based method.

As can be seen, the image quality is improved significantly in Figs. 7(d)–10(d), where all point scatterers are focused well. However, there is still evident residual defocusing in images processed by the classical PGA method and ISAR-based method. These results verify that the proposed NLCS-based method can effectively correct the space-variant phase errors caused by ship rotations, including pitch, roll, and yaw.

B. Real Measured Data

This section verifies the effectiveness of the proposed method by processing real spaceborne SAR images. In order to show the robustness of the NLCS-based refocusing approach, SAR images obtained by three different spaceborne SAR systems are processed, namely, PuJiang-2 (PJ-2), GaoFen-3 (GF-3) [35], and ICEYE [36]. The main system parameters and details of the data sources are listed in Tables IV and V, respectively. The PJ-2 SAR system was operated in the sliding-spotlight mode with 1-m resolution in azimuth, while GF-3 and ICEYE systems were operated in the spotlight mode. Ship targets with defocusing are extracted from SLC images and processed individually. Figs. 11–14 illustrate the processing results on PJ-2 data, Figs. 15–17 show the results on GF-3 data and Figs. 18 and 19 show the results on ICEYE data. SLC images of ships after SAR imaging processing are given in Figs. 11(a)–19(a), in which obviously defocusing can be seen because of ship motions. Figs. 11(b)–19(b) give the images after PGA processing, Figs. 11(c)–19(c) give the images after ISAR-based refocusing, and Figs. 11(d)–19(d) show the results after processed by the NLCS-based refocusing method. A traditional PGA method can only correct phase errors caused by translational motions. Thus, only one or several dominate point scatterers are focused well, while other point scatterers are still blurred due to residual azimuth-variant phase errors. This also indicates that in addition to translational motions, these ship targets also have rotational motions that conventional refocusing methods cannot correct. The ISAR-based method can improve the image quality to some extent, yet the imaging results still have varying degrees of defocusing. Compared with the PGA method and the ISAR-based method, the proposed method can produce better focused images, in which the structure of the ships and the point scatterers can be recovered clearly.

The image entropy, the image contrast, and the peak power of these images are tested for quantitative evaluation of image quality. The image entropy is defined as (36). A low entropy represents a better focused image. The image contrast is defined as [7]

$$IC = \frac{\frac{1}{LM}\sum_{l=0}^{L-1}\sum_{m=0}^{M-1}|I(l,m)|^2}{\left(\frac{1}{LM}\sum_{l=0}^{L-1}\sum_{m=0}^{M-1}|I(l,m)|\right)^2}. \tag{39}$$

A better focused image corresponds to a higher image contrast, with larger differences between bright and dark areas. The peak power reflects the degree of focus of the strongest point scatterer in the image. The numerical evaluation results are listed in Table VI. As the PGA is a scatterer-based refocusing algorithm, the peak power may improve after processing. However, the improvement in entropy and contrast is not significant for rotating targets. The ISAR-based method could increase image entropy and contrast to some extent but sacrifices SNR and leads to a decrease of peak power. Images refocused by the proposed method have almost the best entropy and contrast. Due to avoiding energy loss, the proposed method also ensures that most images have the best peak power. The computational costs are also recorded to show the algorithm efficiency in Table VI. All experiments are conducted on a computer with Intel[1] Core[1] i7-10750H CPU @ 2.60 GHz and 64-GB RAM. Since PGA only corrects the translational motion error, it is much faster than another two refocusing algorithms that further consider the rotational motion error. Our method has almost similar computational costs as the ISAR-based approach, without increasing computational complexity.

The refocusing results and the quantitative assessment both indicate the robustness and effectiveness of the NLCS-based method for refocusing rotating ships in spaceborne SAR imagery.

V. Conclusion

This article presents a novel NLCS-based refocusing method for rotating ships in SAR imagery. First, azimuth-invariant phase error compensation is conducted to remove the effects of translational motions. Then, an equivalent azimuth inverse compression operation without zero-padding operation is designed to recover the azimuth FM of the target image. Finally, the azimuth-variant phase errors induced by target rotations are corrected by the derived high-order NLCS operation, during which the perturbation parameters are estimated via optimizing SAR image entropy. The good performance of the proposed refocusing method has been proven by processing simulated and real spaceborne SAR data. Based on the assumptions of this article, it is reasonable to approximate the rotation of a ship by a third-order polynomial. With the

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increase of synthetic aperture time, higher order polynomial will be required to describe target motions, and higher order NLCS-based correction should be operated for phase error compensation.

The proposed refocusing method performs well on the spaceborne SAR imagery of large ships, such as aircraft carriers, destroyers, cargo ships, and tankers. The rotation of these large ships can be approximately modeled as a high-order polynomial, which can be effectively compensated by the method presented in this article. However, the rotation of smaller ships, such as fishing boats, tends to be more complex with a shorter rotation period. Therefore, the performance of the proposed method might decrease for small rotating ships, particularly when the rotation period is shorter than the synthetic aperture time. One potential solution for this issue is to select appropriate subaperture data for processing. In addition, the proposed method requires the background of the ship image to be the sea surface. When there are strong land targets in the image, such as a ship docked in a harbor, the effectiveness of the proposed method will be affected. The geometric distortion caused by ship rotations is also ignored in this article. The linear phase errors caused by ship movements should be analyzed in detail for the correction of geometric distortion. These challenges will be addressed in future research.

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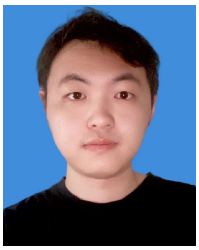
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